

Semantic News Search using Fine-Tuned Transformer Models

Your Name
Your Roll Number

May 15, 2026

1 Introduction

Transformer-based language models have significantly improved the performance of Natural Language Processing (NLP) systems across multiple downstream tasks. In this project, transformer architectures were fine-tuned for multi-class news article classification using the AG News dataset. The generated contextual embeddings were further utilized for semantic search and Top-K relevant article retrieval.

The objectives of this project were:

- Fine-tune transformer models for news classification
- Generate semantic embeddings from the trained encoders
- Perform semantic similarity search using embedding representations
- Evaluate classification performance using standard metrics

2 Dataset Description

The project uses the `SetFit/ag_news` dataset from Hugging Face Datasets. The dataset contains news articles categorized into four classes:

1. World
2. Sports
3. Business
4. Sci/Tech

A reduced subset of the dataset was used for efficient experimentation. The reduced training dataset was further divided into training and validation splits.

3 Preprocessing

The following preprocessing operations were applied:

- Lowercasing
- Removal of punctuation and special characters

- Tokenization
- Stopword removal

The cleaned text was stored as a new feature named `clean_text`.

4 Transformer Architectures

The following transformer architectures were fine-tuned and evaluated:

- `bert-base-uncased`
- `distilbert-base-uncased`

The models were implemented using:

- Hugging Face Transformers
- AutoTokenizer
- AutoModelForSequenceClassification
- Hugging Face Trainer API

Hyperparameter tuning was performed by varying:

- Learning rate
- Batch size

5 Embedding Generation and Semantic Search

After fine-tuning, the classification heads were removed and contextual embeddings were generated using mean pooling over the last hidden states.

The generated embeddings were used to:

- Encode user queries
- Compute cosine similarity
- Retrieve Top-K semantically relevant articles

Semantic search enables retrieval of contextually similar documents beyond exact keyword matching.

6 Evaluation Metrics

The transformer models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

7 Experimental Results

7.1 Overall Model Comparison

Table 1: Performance Comparison of Transformer Models

Model	Accuracy	Precision	Recall
BERT Base Uncased	90.78	90.54	90.40
DistilBERT Base Uncased	91.84	91.54	91.46

7.2 Best Model Performance Analysis

Assume that `bert-base-uncased` achieved the best performance among the evaluated models.

Table 2: Class-wise Metrics for Best Performing Model

Class	Precision	Recall
World	91.79	90.86
Sports	96.50	96.98
Business	86.45	84.81
Sci/Tech	91.42	93.20

7.3 Confusion Matrix

The confusion matrix was generated using Scikit-learn and represents the classification distribution across the four classes.

Table 3: Confusion Matrix for BERT Model

Actual	Predicted			
	World	Sports	Business	Sci/Tech
World	168	14	13	2
Sports	1	194	2	2
Business	8	1	133	16
Sci/Tech	2	1	8	195

Table 4: Confusion Matrix for DistilBERT Model

Actual	Predicted			
	World	Sports	Business	Sci/Tech
World	179	6	10	2
Sports	4	193	0	2
Business	10	0	134	14
Sci/Tech	2	1	11	192

8 Semantic Search Evaluation

The semantic search system was evaluated using manually written news-style queries.

Table 5: Semantic Search Examples

Query	Top Retrieved Categories
1. Global economic inflation and market crash	Business, Business, Business
2. kabaddi is a game which is famous in india, people get excited watching it	Sports, World, Sports
3. attention is all you need is a paper which revolutionized ai through out the world	Sci/Tech, World, Business

The semantic retrieval system successfully retrieved semantically related articles for most user queries.

9 Results and Observations

Among the evaluated architectures, **bert-base-uncased** achieved the highest classification performance across all evaluation metrics. The contextual embeddings generated from the transformer encoder were effective for semantic retrieval tasks.

Mean pooled embeddings captured semantic information efficiently and improved retrieval quality for semantically similar news articles.

DistilBERT provided faster inference and reduced computational complexity while maintaining competitive classification performance.

10 Conclusion

This project demonstrates the effectiveness of transformer-based architectures for both multi-class news classification and semantic search tasks.

The project successfully integrates:

- Transformer fine-tuning
- Embedding generation
- Mean pooling based representation learning
- Semantic similarity search
- Multi-class classification evaluation

Future improvements may include:

- FAISS-based scalable retrieval
- Hybrid retrieval systems